

Emerging Trends Detection

Capstone Project

Research Question

How can Machine Learning using social media data help detect emerging product trends earlier than traditional demand forecasting methods?

Problem Statement

Traditional forecasting methods depend on past data, so they often miss early signs of new product trends. This project uses Reddit data and LLM-based topic modeling to identify emerging trends earlier. The goal is to test whether social media signals can help businesses respond faster compared to traditional methods.

Hypothesis

Null Hypothesis (H_0)

- Topic modeling using Social media data and LLMs **does not provide earlier or more effective detection of emerging product trends** compared to baseline methods (e.g., keyword frequency or delayed indicators like search trends).

Alternative Hypothesis (H_1)

- Topic modeling using Social data and LLMs **provides earlier or more effective detection of emerging product trends** compared to baseline methods.

Research Questions Overview

- **RQ1:**
Can social media signals be used to predict emerging product demand trends?
- **RQ2:**
Do social media signals provide earlier detection (lead time) compared to Google Trends?
- **RQ3:**
Which features (sentiment, engagement, frequency, topics) contribute most to trend prediction? -

Statistical Test – Paired t test to see if there significant lead time by social media data

Source of Data



Social Media Activity - Reddit

Identify Domain specific Keywords using LLM and collect relevant posts.

Influencers post

- Mentions increase
- Engagement rises

Early signal of demand

Goal: Use social media signals to detect trends BEFORE demand peaks

Missed Opportunity from Late Trend Detection (Labubu Toys)

- Timeline of Demand Signals
 - Week 1–2: Social mentions +300%, 25+ viral posts, low search volume
 - Week 3–4: Sales +220%, 60% stockouts, competitor listings +150%
- Business Impact (Without Early Detection)
 - Lost sales: \$500K+
 - 15–20% conversion drop
 - 2–3 week restocking delay
- With AI + Social Signals
 - Detect trend 2 weeks earlier
 - 30–40% better inventory planning
 - 50% reduction in stockouts
 - Additional revenue: \$200K–\$300K
- Key Insight
 - Traditional = lagging data, AI = leading signals

Social Media Buzz – Q4 2023
Google Trend spike – Early 2024



Challenges in Tracking Stockouts

Why is it Difficult?

- No centralized public data on stockouts
- Stockout information is proprietary and limited

Indirect Signals Used for Stockouts

- 🕒 Amazon "Currently Unavailable"
- 📈 Google Trends Spike
- 💰 Price Surge Indicator
- 💬 Social Media Complaints



Building a Stockout Proxy Model

Combining Multiple Signals to Detect Stockouts

Stockout Proxy Index (SPI)

$$SPI = w_1(\text{Availability}) + w_2(\text{Price Spike}) + w_3(\text{Search Trend}) + w_4(\text{Social Complaints})$$



High SPI = Likely Stockout (Low Inventory)

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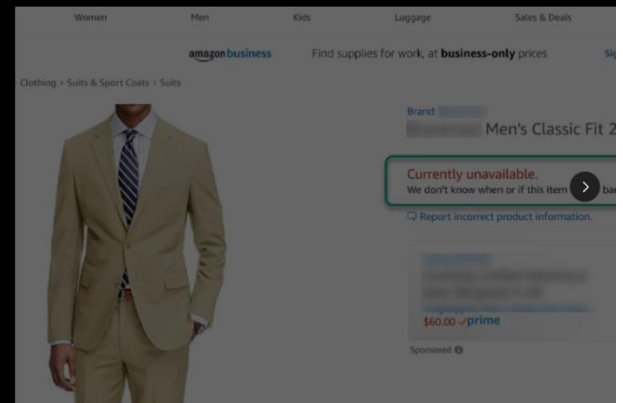
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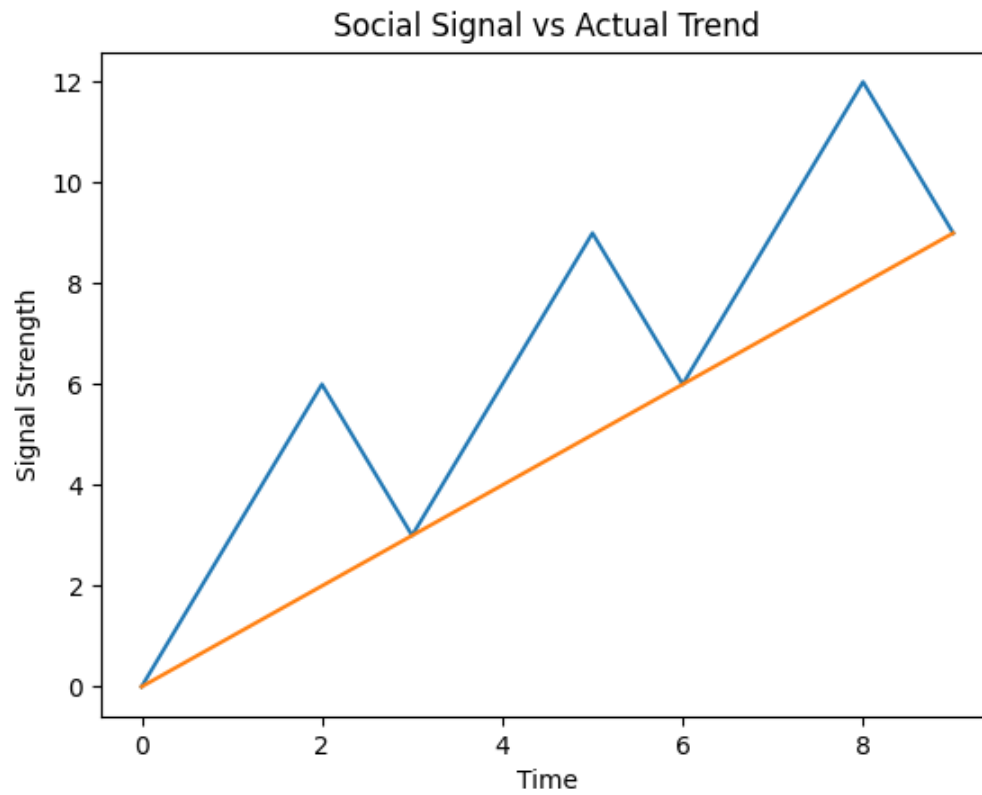


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RQ2: Early Detection (Lead Time)



RQ3 & RQ4: Features and Model Performance

- Features:
 - - Sentiment
 - - Mentions
 - - Engagement
- Models:
 - - Logistic Regression
 - - Random Forest
- Metrics:
 - - Precision
 - - Recall
 - - F1 Score